IRTc: Copula-Based Item Response Theory

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IRTc: Copula-Based Item Response Theory

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About this thesis

This is the landing page for Damian Betebenner's dissertation, "IRTc: Copula-Based Item Response Theory", at dataimago (Independent research — educational measurement & psychometrics (dataimago)).

The thesis is rendered via Quarto + XeLaTeX to a PDF whenever you push changes to `ui/www/` or `R/`. See `docs/thesis.pdf` for the latest build (a stable copy is committed by the `build-thesis.yml` workflow).

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2.2 Formatting

See `THESIS-CLS-README.md` for the 3-mode thesis class file strategy + how to iterate on your `thesis.cls` for institutional compliance.

Chapter 3

Front matter

Abstract

Your dissertation abstract — typically one paragraph, ~250–350 words depending on institutional convention. Pull from your `dataimago-spec.yaml`'s `vertical.dissertation.thesis.contribution` field as a starting point.

Acknowledgments

Acknowledgments for your advisor, committee, family, funders. If you have funding to acknowledge, also fill in your `dataimago-spec.yaml`'s `funding` section — that field can be used to auto-render this section in a future iteration.

Dedication

Optional. Remove this section if not used.

Chapter 4

Introduction

4.1 Background

Conventional IRT treats local independence as the default assumption: conditional on ability, item responses factorize into a product of item-level probabilities. The founding conjecture of this project — treated strictly as a hypothesis, not evidence (claim C0001 in the ledger) — is that if Lord-era IRT had had access to Sklar's theorem, copula theory, and modern compute, parts of IRT might have developed as a dependence-modeling framework in which local independence is simply the independence-copula case rather than the default.

Expand from `dataimago-spec.yaml` □ `vertical.dissertation.thesis.background`*: the intellectual genealogy (student growth percentiles, large-scale assessment, local dependence and testlets as persistent practical realities) and the dual-purpose framing (the dissertation as the first end-to-end stress test of the dataimago research machine).*

4.2 Research question

Can a copula-first formulation of Item Response Theory (IRTc) recover conventional local-independence IRT as the independence-copula special case — and on which real item-response datasets does separating item-response marginals from dependence structures improve held-out prediction, model fit, and diagnostic interpretability relative to Rasch/2PL/3PL/GRM/MIRT baselines?

4.3 Contribution

1. **Theory** — a formally specified IRTc model family: marginals separated from dependence structures, identifiability assumptions for discrete responses stated explicitly, and a precise characterization of which conventional IRT models are special cases, approximations, or mere analogies.
2. **Empirics** — fair, reproducible benchmarks against conventional baselines on simulated and Item Response Warehouse (IRW) data, with failure cases documented as first-class results.
3. **Infrastructure** — a reproducible research application in which every material claim is citation-backed via a claim ledger and every experiment is registered before it runs.

4.4 Roadmap

Chapter 2 frames the historical counterfactual. Chapter 3 examines conventional IRT and the local independence assumption. Chapter 4 introduces copulas and Sklar's theorem. Chapter 5 confronts the discrete response problem. Chapter 6 defines the IRTc model family and locates conventional IRT within it. Chapter 7 develops estimation strategies and simulation studies. Chapter 8 presents IRW benchmark datasets and results. Chapter 9 closes with diagnostics, sensitivity analyses, and limitations.

Chapter 5

Historical Framing: From Empirical Curves and Normal Ogives to Dependence Modeling

5.1 The empirical-curve and normal-ogive lineage

Trace the development from empirical item characteristic curves through normal-ogive and logistic models (Lord, Novick, Rasch, Birnbaum, Samejima, Bock) — primary sources per the WP1 bibliography; every historical claim carries a source ID.

5.2 Local independence as a founding assumption

How and why conditional independence became the default: computational constraints, estimation tractability, and the era's available mathematics.

5.3 The counterfactual, stated carefully

The Lord-era conjecture (claim C0001, $hypothesis$): a generator of research questions, never evidence. What the counterfactual does and does not license — per the claim-ledger rules, this chapter frames questions; it proves nothing.

5.4 Dependence modeling arrives

Sklar (1959) to modern copula families (Joe, Nelsen); factor-copula item response models (Kadhem & Nikoloulopoulos 2021) as the closest existing work.

Chapter 6

Conventional IRT and the Local Independence Assumption

6.1 The conventional model family

Rasch/1PL, 2PL, 3PL, graded response, (generalized) partial credit, MIRT and bi-factor — the Level-0 baselines of the model ladder, implemented via `mirt` and `TAM`.

6.2 Local independence, formally

For a response vector $Y_i = (Y_{i1}, \dots, Y_{iJ})$:

$$P(Y_i = y_i \mid \theta_i) = \prod_j P(Y_{ij} = y_{ij} \mid \theta_i, \xi_j)$$

State the assumption precisely; distinguish it from unidimensionality.

6.3 Where the assumption strains

Local dependence, testlets, item bundles, speededness, multidimensionality — the persistent practical realities conventional models treat as nuisances. Q3-style residual diagnos-

tics and limited-information fit statistics as the detection toolkit.

6.4 Existing accommodations

Testlet response theory, bi-factor/two-tier models, latent class IRT — what each concedes to dependence, and at what interpretive cost.

Chapter 7

Copulas, Sklar's Theorem, and Dependence Structures

7.1 Copulas and Sklar's theorem

Definitions, Sklar's theorem, and the uniqueness caveat: uniqueness holds under continuous marginals — a caveat that becomes central in Chapter 5.

7.2 Copula families and their dependence signatures

Gaussian/BVN, Student t, Frank, Clayton/Gumbel and rotations, Joe/BB families: tail dependence, asymmetry, and what each signature could mean psychometrically (items jointly failing among low-ability examinees; region-specific discrimination).

7.3 Structured constructions

Factor copulas, bi-factor and second-order copulas, vines, factor trees — the constructions the model ladder's Levels 2–4 build on (FactorCopula as the implementation reference; capability claims only).

Chapter 8

The Discrete Response Problem

The load-bearing chapter for mathematical honesty — see `wiki/theory/discrete-copula-cautions.m` *written before any model claims.*

8.1 Why discreteness breaks naive Sklar arguments

Item responses are dichotomous or ordinal; the copula linking observed responses is not unique (claim C0002). No "mathematically cleaner" claim may lean on continuous-Sklar uniqueness.

8.2 Candidate resolutions

1. **Latent-continuous formulation** — categories as thresholds of latent continuous response variables; copulas on the latent scale.
2. **Exact rectangle-probability likelihoods** — category probabilities as copula rectangle differences.
3. **Bayesian data augmentation** — Smith & Khaled (2012) as the anchor caution: estimation difficulty beyond the bivariate case.
4. **Parametric factor-copula constructions** — existing discrete-data factor copula models as the starting point.
5. **Distributional/randomized transforms** — only if the literature supports them for the target inference.

8.3 Identifiability conditions

State what is and is not identified for dichotomous and ordinal data under each resolution — every theorem-like statement gets proof, citation, or a `conjecture` label (WP2 gates).

Chapter 9

The IRTc Model Family and Conventional IRT as Special Cases

9.1 The IRTc construction

$$P(Y_i \le y_i \mid \theta_i) = C_\theta\big(F_1(y_{i1} \mid \theta_i, \xi_1), \dots, F_J(y_{iJ} \mid \theta_i, \xi_J);\ \delta\big)$$

Item-response marginals F_j separated from the dependence structure C_θ with parameters δ; copula placement options (among observed responses conditional on θ, among latent response variables, among latent dimensions, among residuals, among testlet effects).

9.2 Conventional IRT located within IRTc

The independence copula recovers local-independence IRT (Level 1 of the model ladder — the product-copula wrapper with its numerical-tolerance acceptance test). For each conventional model, classify the relationship: exact equivalence, approximation, or analogy — nothing stronger than what is proved.

9.3 Parameter interpretation under dependence

What difficulty, discrimination, guessing, and thresholds mean when dependence is no longer Gaussian or product-form; when a copula parameter is a substantive estimand vs a nuisance parameter for fit; whether any Rasch-style invariance survives restricted copula families.

Chapter 10

Estimation Strategies and Simulation Studies

10.1 Estimation strategies

Per model-ladder level: `mirt`/`TAM` baselines (EM, MHRM); FactorCopula estimation for one-factor and structured copulas; rectangle likelihoods; Bayesian augmentation where justified. Every fit records versions, seeds, optimizer settings, convergence diagnostics, and warnings (WP3 gates — no silent inclusion of nonconverged fits).

10.2 Simulation designs

Every model has a simulation recipe before touching real data (WP2 gate): binary 2PL and ordinal GRM generators first, then copula-family generators for parameter recovery.

10.3 Parameter recovery

Recovery RMSE and bias by copula family, sample size, and item count; stability across optimization starts.

10.4 The Level-1 equivalence test

The product-copula wrapper must match baseline estimates, log-likelihoods, and predicted probabilities within numerical tolerance — differences explained and documented, or the ladder does not advance.

Chapter 11

IRW Benchmark Datasets and Results

11.1 The Item Response Warehouse

IRW v28.2: 900+ harmonized open item-response datasets, 2B+ responses (Behavior Research Methods, 2025). Access via the irw R package with Redivis authentication. Not every IRW dataset suits every IRTc model — selection criteria and data cards per WP4.

11.2 Benchmark design

Tiered datasets (smoke-test / primary / stress-test); immutable raw zone; reproducible splits; fairness rules from charter §14.3 — same preprocessing, same splits, same inclusion criteria, failed fits reported, stronger baselines before any IRTc gain is claimed.

11.3 Results

Held-out log loss and Brier score as primary metrics; marginal likelihood, AIC/BIC/SABIC, M2, calibration by ability strata as secondary. Every table carries model version, dataset version, split ID, and seed. Results appear here only from pre-registered experiments ($experiments/registry.yaml$).

11.4 Failure cases

Documented as first-class results — datasets where IRTc does not improve on baselines are reported with the same prominence as successes.

Chapter 12

Diagnostics, Sensitivity Analyses, and Limitations

12.1 Diagnostics

Residual-dependence (Q3-style) reduction, tail-dependence diagnostics, local-dependence heatmaps, calibration curves, ability-estimate comparisons — can IRTc diagnostics see what conventional residual diagnostics miss?

12.2 Sensitivity analyses

Copula family set, starting values, quadrature/integration method, sample and item subsampling, missingness handling, category collapsing, group/testlet definitions, split construction, optimizer thresholds (charter §14.4). Sensitivity precedes any superiority claim (WP5 gate).

12.3 Interpretability and stability

Are the additional dependence parameters stable and interpretable at realistic educational-testing sample sizes, or does model-selection risk erase the gains?

12.4 Limitations

Required by the claim-ledger rules: where IRTc is not yet justified, where identifiability remains conjectural, where compute constrains the ladder, and what the acceptance criteria (charter §20) still leave open.

Chapter 13

Back matter

References

The bibliography is rendered automatically from `../references.bib`. No content needed here — Quarto handles it.

Appendices

Optional. If you have appendices (data tables, supplementary analyses, code listings, interview protocols), add them as `chapters/99a-appendix-X.qmd`, `99b-appendix-Y.qmd`, etc. and reference them in `../_quarto.yml`.